

→ Formalize O, Ω

Best case / worst case

$f(n)$ is $O(g(n))$ if for sufficiently large n $f(n)$ is upper bounded by a constant multiple of $g(n)$.

$f(n)$ is $O(g(n))$ if $\exists$ $n_0 \geq 0, c > 0$ such that $\forall n \geq n_0 : f(n) \leq c \cdot g(n)$

Annotations:
"here exist" points to $\exists$
"for all" points to $\forall$

claim $4n+12$ is $O(n^2)$

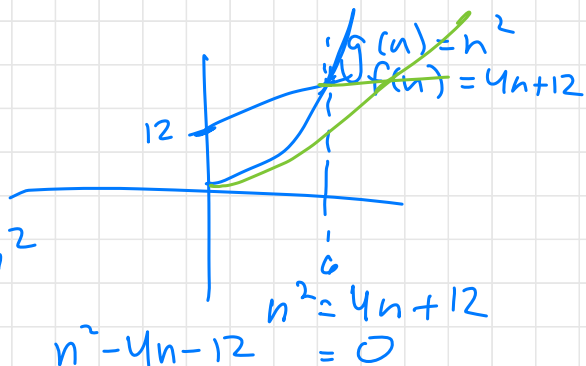
Proof To show that $4n+12$ is $O(n^2)$, we need to show that $\exists n_0 \geq 0, c > 0$, such that $\forall n \geq n_0 : 4n+12 \leq c \cdot n^2$

consider

$$c = 1$$

$$n_0 = 6$$

$$\forall n \geq 6 : 4n+12 \leq c n^2$$



$$(n+2)(n-6) = 0$$

$$n^2 @ 6 : 36$$

$$4n+12 \quad 12+24=36$$

$$n=6$$

P2 bry o pf practice

false

Claim n^2 is $O(4n+12)$

Proof attempt

we need $n_0 \geq 0$, $C > 0$ s.t.

$$\rightarrow \forall n \geq n_0 : \underbrace{n^2}_{\text{LHS}} \leq \underbrace{C \cdot (4n+12)}_{\text{RHS}}$$

$$C = 10^6$$

consider $n = 10^{12}$

$$\text{LHS } n^2 = (10^{12})^2 = 10^{24}$$

$$\text{RHS } C \cdot (4n+12) = 10^6 (4 \cdot 10^{12} + 12) = 10^6 \cdot 10^{12} = 10^{18}$$

claim n^2 is not $O(4n+12)$

proof show that there do not exist $n_0 \geq 0$, $C > 0$ so that

$$\forall n \geq n_0 : \underbrace{n^2}_{\text{LHS}} \leq \underbrace{C \cdot (4n+12)}_{\text{RHS}}$$

Properties of O

- constants don't matter
- drop lower order terms
- bases of logs don't matter HW
- bases of exponents do matter

Claim if $f(n) = O(g(n) + h(n))$,
then $f(n) = O(\max(g(n), h(n)))$

ex $n^2 + n = O(\overset{g(n)}{n^2} + \overset{h(n)}{n}) \quad f(n) = O(f(n))$
 $\rightarrow n^2 + n = O(\max(n^2, n))$
 $n^2 + n = O(n^2)$

Pf let $f(n) = O(g(n) + h(n))$. (*)

WTS (want to show) that $f(n) = O(\max(g(n), h(n)))$,
so we need $n_0 \geq 0, c > 0$ s.t.

$\forall n \geq n_0 : f(n) \leq c \cdot \max(g(n), h(n))$

We know (*). This means that $\exists n_0' \geq 0, c' > 0$
such that

$\forall n \geq n_0' : f(n) \leq c' \cdot (g(n) + h(n))$

$$f(n) \leq c' \cdot (\max(g(n), h(n)) + \max(g(n), h(n)))$$

$$\forall n \geq n_0 : f(n) \leq c' \cdot 2 \cdot \max(g(n), h(n))$$

So choose $n_0 = n_0'$, $c = 2c'$

$$f(n) = O(\max(g(n), h(n)))$$

Best / worst case?

Alg 1 (int array A of length n)

if $A[1] \geq 0$:

total = 0

for $i = 1, \dots, n$

total = total + $A[i]$

else:

total = 5

$f(n)$ s.t. # prim. ops done by Alg 1 on input of size n is $\Theta(f(n))$

n	$?$
1	$.$

Worst-case runtime:

max # steps needed on input
size n